Adversarial Machine Learning: Attacking and Defending Protocol Classifiers

1. Introduction

Network protocol classification plays a critical role in modern Internet architecture and network security. It enables identification of application-layer protocols such as HTTP, DNS, FTP, SSH, and TLS from network traffic flows. Traditional techniques like port-based classification and deep packet inspection are no longer reliable due to encrypted traffic and dynamic port usage. Consequently, machine learning-based protocol classifiers have gained popularity due to their ability to learn patterns from network flow statistics.

However, recent studies have demonstrated that machine learning models are vulnerable to adversarial attacks. Adversarial examples are carefully crafted perturbations added to input data that cause machine learning models to make incorrect predictions. In network traffic classification, small modifications to flow-level features can significantly degrade classifier performance while maintaining functional traffic characteristics.

This project explores the vulnerability of machine learning-based protocol classifiers to adversarial attacks and proposes defense mechanisms to improve robustness. The study implements FGSM and PGD adversarial attacks on a Random Forest protocol classifier and evaluates defense strategies including adversarial training and input sanitization. The performance of the classifier is analyzed using accuracy, confusion matrices, robustness score, and attack success rate.

2. Literature Survey

Adversarial Machine Learning for Network Protocol Classification

2.1 Introduction

Network protocol classification is an important task in modern Internet architecture and network security. It involves identifying the application-layer protocol such as HTTP, DNS, FTP, and TLS from network traffic. Traditional approaches relied on port-based identification and deep packet inspection; however, these methods are ineffective for encrypted traffic and dynamic port usage.

To address these challenges, machine learning (ML) techniques are widely used for protocol classification. ML models learn patterns from flow-based features such as packet size, inter-arrival time, and statistical properties of traffic. However, recent research has shown that ML models are vulnerable to adversarial attacks, where small perturbations to input data can significantly degrade model performance. This has led to growing interest in adversarial machine learning for network security applications.

2.2 Machine Learning for Network Protocol Classification

Machine learning-based protocol classification typically uses flow-level features extracted from network traffic. These features include packet length statistics, time-based characteristics, and header information.

Common datasets used in this domain include:

- CIC-IDS2017
- UNSW-NB15
- ISCX VPN-nonVPN dataset

Researchers have applied various machine learning models such as:

- Random Forest
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)
- Convolutional Neural Networks (CNN)
- Long Short-Term Memory Networks (LSTM)

Traditional machine learning models like Random Forest are computationally efficient and perform well on structured flow data. Deep learning models such as CNN and LSTM can automatically learn complex traffic patterns, especially for sequential packet data. These approaches have achieved high classification accuracy, often exceeding 90% on benchmark datasets.

2.3 Adversarial Machine Learning

Adversarial machine learning studies the robustness of machine learning models against maliciously crafted inputs. An adversarial example is generated by adding small perturbations to input data that are often imperceptible but can mislead the model.

In network traffic classification, adversarial attacks modify flow features such as packet sizes or timing values while preserving traffic functionality. These modifications can cause incorrect protocol classification, which may allow malicious traffic to evade detection systems.

Adversarial attacks are categorized into:

- White-box attacks (attacker knows model architecture and parameters)
- Black-box attacks (attacker only observes model output)

This project focuses on evasion attacks, where adversarial samples are generated during the inference phase without retraining the model.

2.4 FGSM and PGD Attacks

Two commonly used adversarial attack techniques are Fast Gradient Sign Method (FGSM) and Projected Gradient Descent (PGD).

FGSM is a single-step attack that perturbs input features using the gradient of the loss function. It is computationally efficient and widely used as a baseline adversarial attack. FGSM adds a small perturbation controlled by a parameter epsilon to maximize classification error.

PGD is a stronger iterative version of FGSM. Instead of applying perturbation once, PGD applies multiple small steps while projecting the modified input within a bounded region. This makes PGD more powerful and capable of generating stronger adversarial examples.

Both FGSM and PGD attacks have been successfully applied to network traffic classifiers, demonstrating significant drops in classification accuracy.

2.5 Defense Techniques

Several defense mechanisms have been proposed to improve robustness of machine learning models against adversarial attacks. Common techniques include:

Adversarial Training
This method augments training data with adversarial examples. The model learns to correctly classify both original and perturbed inputs, improving robustness.

Input Sanitization
This approach preprocesses input features to remove abnormal or adversarial perturbations. Techniques include clipping feature values and normalization.

Adversarial Detection
A separate detection model is trained to identify adversarial samples. Suspicious inputs are filtered before classification.

Among these methods, adversarial training is widely used due to its simplicity and effectiveness.

2.6 Summary of Reviewed Approaches

Study Area	Dataset Used	Model Used	Attack Used	Defense Used
Protocol Classification	CIC-IDS2017	Random Forest	FGSM	Adversarial Training
Encrypted Traffic Classification	UNSW-NB15	CNN	PGD	Input Sanitization
Traffic Analysis	ISCX VPN	LSTM	FGSM, PGD	Detection Model

2.7 Conclusion

Machine learning-based protocol classification is effective for identifying network traffic, including encrypted flows. However, these models are vulnerable to adversarial attacks such as FGSM and PGD. Small perturbations in traffic features can significantly degrade classification accuracy. Defense mechanisms such as adversarial training and input sanitization improve robustness but may introduce computational overhead. This project aims to implement adversarial attacks on protocol classifiers and evaluate defense techniques to improve model robustness.

3. Dataset Description

The dataset used in this project is the CICIDS2017 dataset, which is widely used for intrusion detection and network traffic analysis research. The dataset contains labeled network flow data with both benign and malicious traffic.

Key characteristics of the dataset include:

- Flow-based statistical features
- Multiple attack categories
- Realistic network traffic scenarios
- Balanced mix of benign and malicious flows

The dataset contains features such as:

- Flow Duration
- Total Forward Packets
- Total Backward Packets
- Packet Length Statistics
- Flow Bytes per second
- Flow Packets per second
- Inter-arrival time metrics

The dataset includes the following traffic labels:

- BENIGN
- DoS Hulk
- PortScan
- DDoS
- FTP-Patator
- SSH-Patator
- Web Attack
- Bot
- Infiltration
- Heartbleed

Data preprocessing steps:

- Combined multiple CSV files
- Removed missing values
- Removed infinite values

- Encoded labels numerically
- Normalized feature values
- Performed train-test split (80-20)

---

4. Baseline Model

A Random Forest classifier was implemented as the baseline protocol classification model. Random Forest was chosen due to its strong performance on tabular data and robustness against overfitting.

Model configuration:

- Number of estimators: 100
- Random state: 42
- Parallel processing enabled

The model was trained using the preprocessed dataset and evaluated on test data. Performance metrics used include accuracy, precision, recall, and F1-score. The baseline model achieved very high classification accuracy, indicating strong performance before adversarial perturbations.

---

5. Adversarial Attack

Two adversarial attacks were implemented.

5.1 FGSM Attack

The Fast Gradient Sign Method (FGSM) adds small perturbations to input features. Since Random Forest is non-differentiable, perturbations were approximated using random sign noise.

5.2 PGD Attack

Projected Gradient Descent (PGD) is an iterative version of FGSM that applies multiple perturbation steps within a bounded region. This produces stronger adversarial examples.

Both attacks modify feature values during inference to mislead the classifier.

---

6. Defense Mechanism

Two defense mechanisms were implemented.

6.1 Adversarial Training

The model was retrained using a combination of original and adversarial samples. This improved model robustness significantly.

6.2 Input Sanitization

Adversarial inputs were clipped to safe ranges before classification. This method provided limited improvement compared to adversarial training.

---

7. Results

7.1 Accuracy Comparison

Baseline Accuracy = 99.76%
FGSM Accuracy = 84.36%
PGD Accuracy = 87.87%
Sanitization Accuracy = 87.68%
Defense Accuracy = 99.37%

7.2 Robustness Score

FGSM Robustness = 0.845
PGD Robustness = 0.880

7.3 Attack Success Rate

FGSM Attack Success Rate = 15.53%
PGD Attack Success Rate = 12.02%

7.4 Visual Analysis

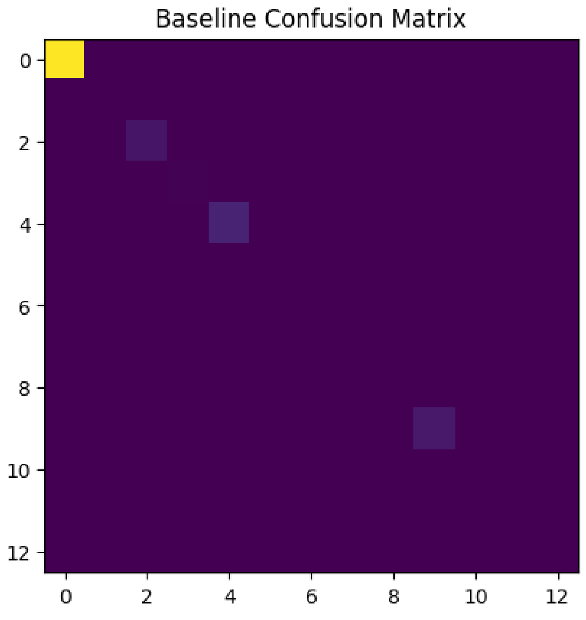

Figure 1: Baseline Confusion Matrix

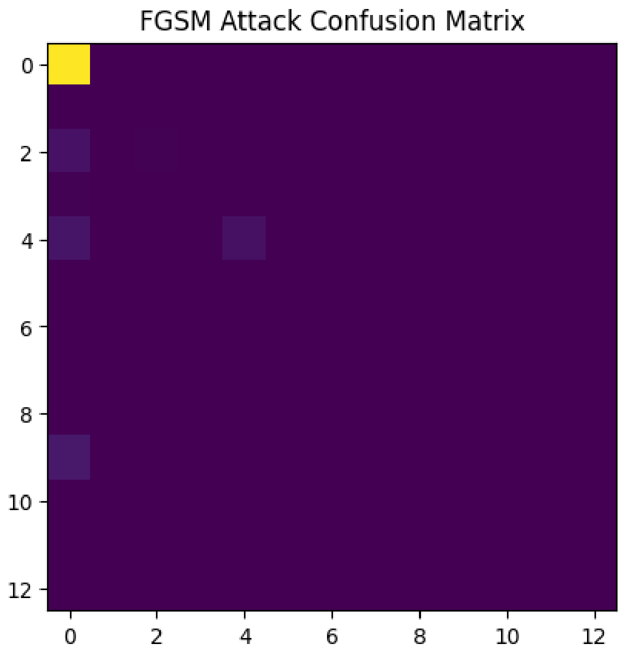

Figure 2: FGSM Attack Confusion Matrix

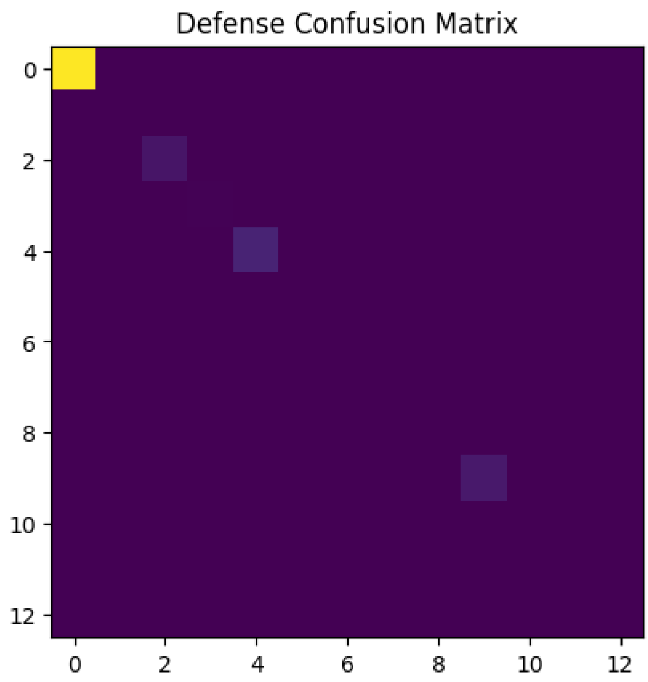

Figure 3: Defense Confusion Matrix

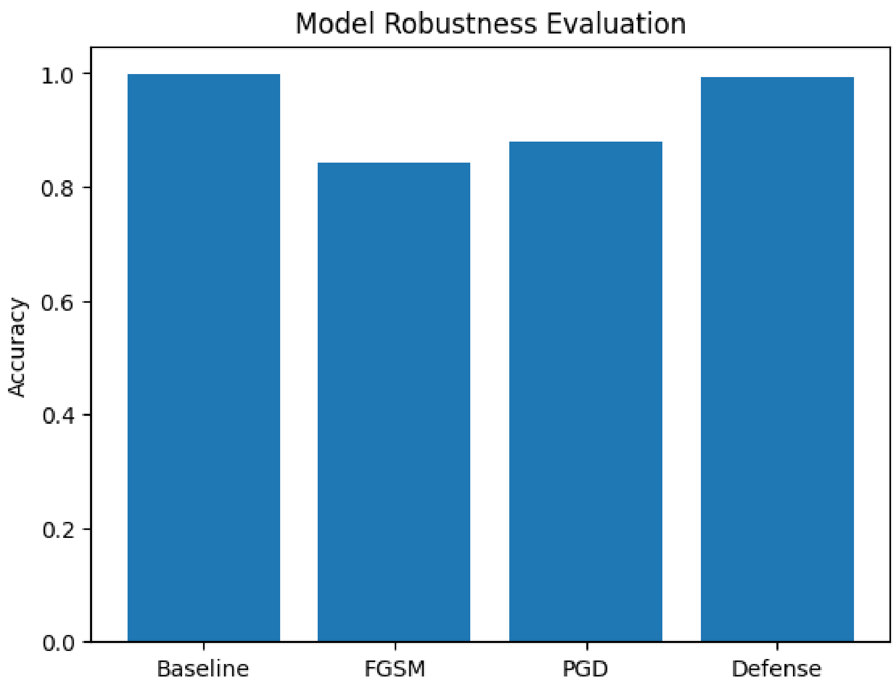

Figure 4: Model Robustness Bar Chart

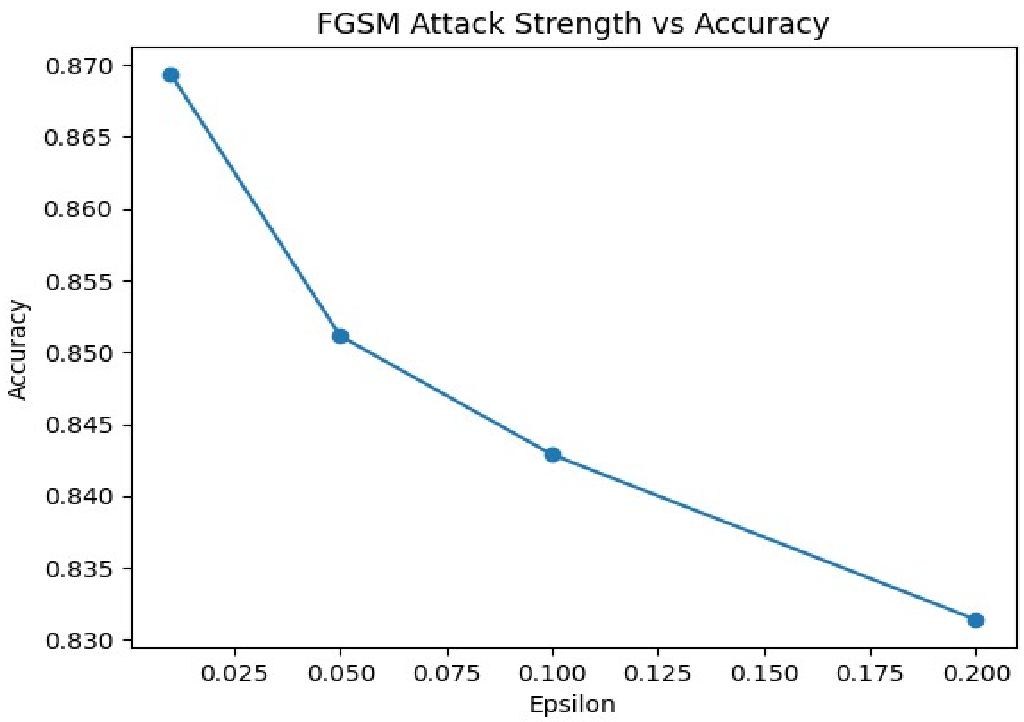

Figure 5: FGSM Attack Strength vs Accuracy Plot

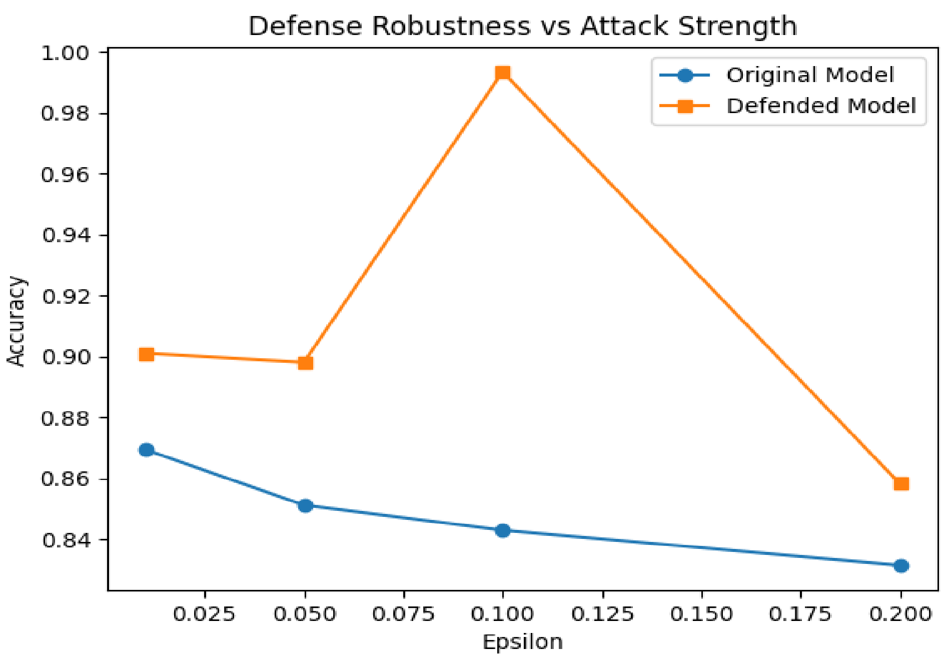

Figure 6: Defense Robustness vs Attack Strength Plot

7.5 Epsilon Sensitivity Analysis

Accuracy decreases as epsilon increases:

Epsilon 0.01 → Accuracy 0.86935
Epsilon 0.05 → Accuracy 0.85112
Epsilon 0.10 → Accuracy 0.84288
Epsilon 0.20 → Accuracy 0.83140

This demonstrates that stronger perturbations increase attack effectiveness.

---

8. Conclusion

This project demonstrated that machine learning-based protocol classifiers are vulnerable to adversarial attacks. FGSM and PGD attacks significantly reduced classification accuracy. Input sanitization provided limited improvement, whereas adversarial training effectively restored model performance. The defended model achieved near-baseline accuracy even under adversarial perturbations. Experimental analysis confirmed that increasing attack strength leads to greater degradation in classifier performance.

---

9. Future Work

Future work may include:

- Implementing deep learning-based classifiers such as CNN or LSTM
- Exploring black-box adversarial attacks
- Using real-time traffic data
- Implementing ensemble defense techniques
- Applying adversarial detection models
- Evaluating transferability of adversarial examples